

SQL Views Exposed to Power BI

Project: AI-Augmented Executive Analytics Platform

Purpose: Document the exact SQL views used as the curated semantic layer for Power BI, including purpose, grain, source tables, key columns, transformations, and model notes.

View Inventory

View Name	Type	Business Purpose	Grain	Primary Consumers
dbo.vw_DimDate	Dimension	Canonical date table for time intelligence	1 row per calendar date	All report pages
dbo.vw_DimCustomer	Dimension	Curated customer profile and geography enrichment	1 row per customer	Segmentation, cohorts
dbo.vw_DimProduct	Dimension	Product hierarchy with category and subcategory	1 row per product	Product and budget analysis
dbo.vw_DimSalesTerritory	Dimension	Geographic hierarchy for regional analysis	1 row per sales territory	Regional views, targets
dbo.vw_FactInternetSales	Fact	Actual sales transactions for KPI, margin and trend analysis	1 row per sales transaction line	Executive KPIs, product, customer
dbo.vw_CubeCustomerCohort	Analytical fact	Pre-aggregated monthly cohort retention and customer status cube	1 row per cohort x month x grouping set	Cohort and loyalty pages

Model Notes

- These views form the curated SQL layer consumed by Power BI. Business-friendly names and derived attributes are pushed into SQL to keep the semantic model thin and easier to govern.
- dbo.vw_DimDate should be marked as the official Date Table in Power BI.
- dbo.vw_FactInternetSales is filtered to the last three years of data to reduce model size and improve refresh performance.
- dbo.vw_CubeCustomerCohort is an analytical view intended for cohort and retention reporting. It is not a transaction fact and should be related to the model carefully, typically through Date and category/country reporting logic rather than as a replacement for the sales fact.

dbo.vw_DimDate

Purpose: Provides a cleaned, reporting-friendly date dimension for all time-intelligence calculations.

Grain: One row per calendar date.

Source: AdventureWorksDW2022.dbo.DimDate

Key Columns: DateKey, Date, Year, Quarter, Month, MonthNo, Start of Month

Transformations: Renames columns to friendly names; derives short labels, Year-Month labels, quarter labels, and start-of-month date; filters to the last three calendar years.

Power BI Notes: Mark this as the Date Table using the [Date] column. Use DateKey for fact relationships and [Date] for DAX time intelligence.

SQL Definition (abridged / formatted)

```
CREATE VIEW dbo.vw_DimDate AS
SELECT
    [DateKey],
    [FullDateAlternateKey] AS [Date],
    [DayNumberOfWeek] AS [DayNo],
    [EnglishDayNameOfWeek] AS [DayName],
    LEFT([EnglishDayNameOfWeek], 3) AS [DayShort],
    [WeekNumberOfYear] AS [WeekNo],
    [MonthNumberOfYear] AS [MonthNo],
    [EnglishMonthName] AS [Month],
    LEFT([EnglishMonthName], 3) AS [MonthShort],
    DATEFROMPARTS([CalendarYear], [MonthNumberOfYear], 1) AS [Start of Month],
    DATEFROMPARTS([CalendarYear], [MonthNumberOfYear], 1) AS [Year-Month],
    CONCAT(LEFT([EnglishMonthName], 3), ' ', [CalendarYear]) AS [Year_Month_Eng],
    [CalendarQuarter] AS [Quarter],
    CONCAT('Q', [CalendarQuarter]) AS [QuarterName],
    CONCAT('Q', [CalendarQuarter], '-', [CalendarYear]) AS [Quarter-Year],
    [CalendarYear] AS [Year],
    [FiscalQuarter],
    [FiscalYear],
    [FiscalSemester]
FROM [AdventureWorksDW2022].[dbo].[DimDate]
WHERE [CalendarYear] >= YEAR(GETDATE()) - 3;
```

Documentation note: The SQL text above is formatted for readability. The cohort view is intentionally summarized here because its full definition is long and already separately maintained in source control.

dbo.vw_DimCustomer

Purpose: Creates a customer dimension enriched with geography and simplified demographic labels for segmentation and cohort analysis.

Grain: One row per customer.

Source: DimCustomer joined to DimGeography and DimSalesTerritory

Key Columns: CustomerKey, Full Name, Gender, AnnualIncome, EducationLevel, DateFirstPurchase, Country, Region

Transformations: Combines first and last name, standardizes gender labels, renames income and geography fields, and enriches customer records with city, state, country, and region.

Power BI Notes: Use CustomerKey as the active relationship to the sales fact. DateFirstPurchase supports cohort and loyalty views.

SQL Definition (abridged / formatted)

```
CREATE VIEW dbo.vw_DimCustomer AS
SELECT TOP 100 PERCENT
    c.CustomerKey,
    c.FirstName + ' ' + c.LastName AS [Full Name],
    c.BirthDate,
    CASE c.gender WHEN 'M' THEN 'Male' WHEN 'F' THEN 'Female' END AS Gender,
    c.YearlyIncome AS AnnualIncome,
    c.TotalChildren,
    c.EnglishEducation AS EducationLevel,
    c.HouseOwnerFlag AS HomeOwner,
    c.datefirstpurchase AS DateFirstPurchase,
    g.city AS [City],
    g.stateprovincename AS [State-Province],
    st.SalesTerritoryCountry AS Country,
    st.SalesTerritoryRegion AS Region,
    g.postalcode AS [Postal Code]
FROM DimCustomer as c
LEFT JOIN DimGeography AS g ON g.geographykey = c.geographykey
LEFT JOIN DimSalesTerritory AS st ON g.SalesTerritoryKey = st.SalesTerritoryKey
ORDER BY CustomerKey ASC;
```

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dbo.vw_DimProduct

Purpose: Creates a curated product dimension with hierarchy and presentation-friendly attributes.

Grain: One row per product.

Source: DimProduct joined to DimProductSubcategory and DimProductCategory

Key Columns: ProductKey, Product SKU, Product Name, Category, SubCategory, Standard Cost, List Price

Transformations: Renames product fields for reporting, adds category and subcategory hierarchy, defaults missing status to Outdated, and exposes product photo for rich reporting if needed.

Power BI Notes: Use ProductKey to join to the sales fact. Category is especially important for budget comparisons and cohort reporting slices.

SQL Definition (abridged / formatted)

```
CREATE VIEW dbo.vw_DimProduct AS
SELECT TOP 100 PERCENT
    p.[ProductKey],
    p.[ProductAlternateKey] AS [Product SKU],
    p.[EnglishProductName] AS [Product Name],
    p.StandardCost AS [Standard Cost],
    p.[Color] AS [Product Color],
    p.ListPrice AS [List Price],
    p.Style AS [Product Style],
    p.[ModelName] AS [Product Model],
    p.ProductSubcategoryKey,
    ps.EnglishProductSubcategoryName AS SubCategory,
    ps.ProductCategoryKey,
    pc.EnglishProductCategoryName AS Category,
    p.LargePhoto AS [Product Photo],
    ISNULL(p.Status, 'Outdated') AS [Product Status]
FROM DimProduct as p
LEFT JOIN DimProductSubcategory AS ps ON ps.ProductSubcategoryKey = p.ProductSubcategoryKey
LEFT JOIN DimProductCategory AS pc ON ps.ProductCategoryKey = pc.ProductCategoryKey
ORDER BY p.ProductKey asc;
```

Documentation note: The SQL text above is formatted for readability. The cohort view is intentionally summarized here because its full definition is long and already separately maintained in source control.

dbo.vw_DimSalesTerritory

Purpose: Provides a simplified sales territory hierarchy for regional analysis.

Grain: One row per sales territory.

Source: DimSalesTerritory

Key Columns: SalesTerritoryKey, Region, Country, Group

Transformations: Renames fields to business-friendly names and excludes records where country equals NA.

Power BI Notes: Use SalesTerritoryKey as the primary relationship to the sales fact. Country and Group support regional and target reporting.

SQL Definition (abridged / formatted)

```
CREATE VIEW dbo.vw_DimSalesTerritory AS
SELECT
    [SalesTerritoryKey],
    [SalesTerritoryRegion] AS Region,
    [SalesTerritoryCountry] AS Country,
    [SalesTerritoryGroup] AS [Group]
FROM DimSalesTerritory
WHERE SalesTerritoryCountry <> 'NA';
```

Documentation note: The SQL text above is formatted for readability. The cohort view is intentionally summarized here because its full definition is long and already separately maintained in source control.

dbo.vw_FactInternetSales

Purpose: Exposes the actual sales fact used for revenue, gross profit, pricing, quantity, and order analysis.

Grain: One row per sales transaction line.

Source: FactInternetSales

Key Columns: ProductKey, OrderDateKey, CustomerKey, SalesTerritoryKey, SalesOrderNumber, Order Quantity, Unit Price, Sales Amount, Total Product Cost, GrossProfit

Transformations: Renames numeric fields for reporting, computes GrossProfit in SQL, and limits the dataset to the last three years based on OrderDateKey.

Power BI Notes: This is the core actuals fact. Keep DAX measures aggregation-based and derive ratios from base measures rather than calculated columns.

SQL Definition (abridged / formatted)

```
CREATE VIEW dbo.vw_FactInternetSales AS
SELECT TOP 100 PERCENT
    [ProductKey],
    [OrderDateKey],
    [DueDateKey],
    [ShipDateKey],
    [CustomerKey],
    [SalesTerritoryKey],
    [SalesOrderNumber],
    [OrderQuantity] AS [Order Quantity],
    [UnitPrice] AS [Unit Price],
    [TotalProductCost] AS [Total Product Cost],
    [SalesAmount] AS [Sales Amount],
    [SalesAmount] - [TotalProductCost] AS GrossProfit
FROM FactInternetSales
WHERE LEFT(OrderDateKey, 4) >= YEAR(GETDATE()) - 3
ORDER BY OrderDateKey ASC;
```

Documentation note: The SQL text above is formatted for readability. The cohort view is intentionally summarized here because its full definition is long and already separately maintained in source control.

dbo.vw_CubeCustomerCohort

Purpose: Provides a pre-aggregated monthly cohort retention cube with customer status transitions and revenue, optimized for Power BI cohort analysis.

Grain: One row per cohort month x activity month x grouping set (category-country, category-only, or grand total).

Source: vw_FactInternetSales joined to vw_DimDate, vw_DimCustomer, and vw_DimProduct via multiple CTEs

Key Columns: cohort_month, cohort_year, month_number, month_num_lab, product_category, country, total_customers_month, retained_customers_month, total_revenue, percentage_month

Transformations: Anchors a rolling 24-month window to the latest transaction month, classifies customer monthly status as New, Returning, Recovered, or Lost, applies grouping sets for subtotals, and hard-guards month 0 retention to 1.0000.

Power BI Notes: Best used for the cohort and loyalty pages only. Because it is already aggregated, avoid joining it back as if it were a transaction fact. Use it as a reporting cube or dedicated analytical fact.

SQL Definition (abridged / formatted)

```
CREATE VIEW dbo.vw_CubeCustomerCohort AS
-- Pre-aggregated monthly cohort retention view with customer status and revenue
-- Main output columns:
-- cohort_month, cohort_year, month_number, month_num_lab, product_category,
-- country, total_customers_month, retained_customers_month, total_revenue,
-- new_customers, returning_customers, recovered_customers, lost_customers,
-- percentage_month
-- Built through CTEs: window_anchor, lost_grace_months, customer_cohort,
-- customer_revenue, customer_activities, customer_monthly_flags,
-- customer_status_classified, retention_aggregated, cohort_size
SELECT ...
;
```

Documentation note: The SQL text above is formatted for readability. The cohort view is intentionally summarized here because its full definition is long and already separately maintained in source control.